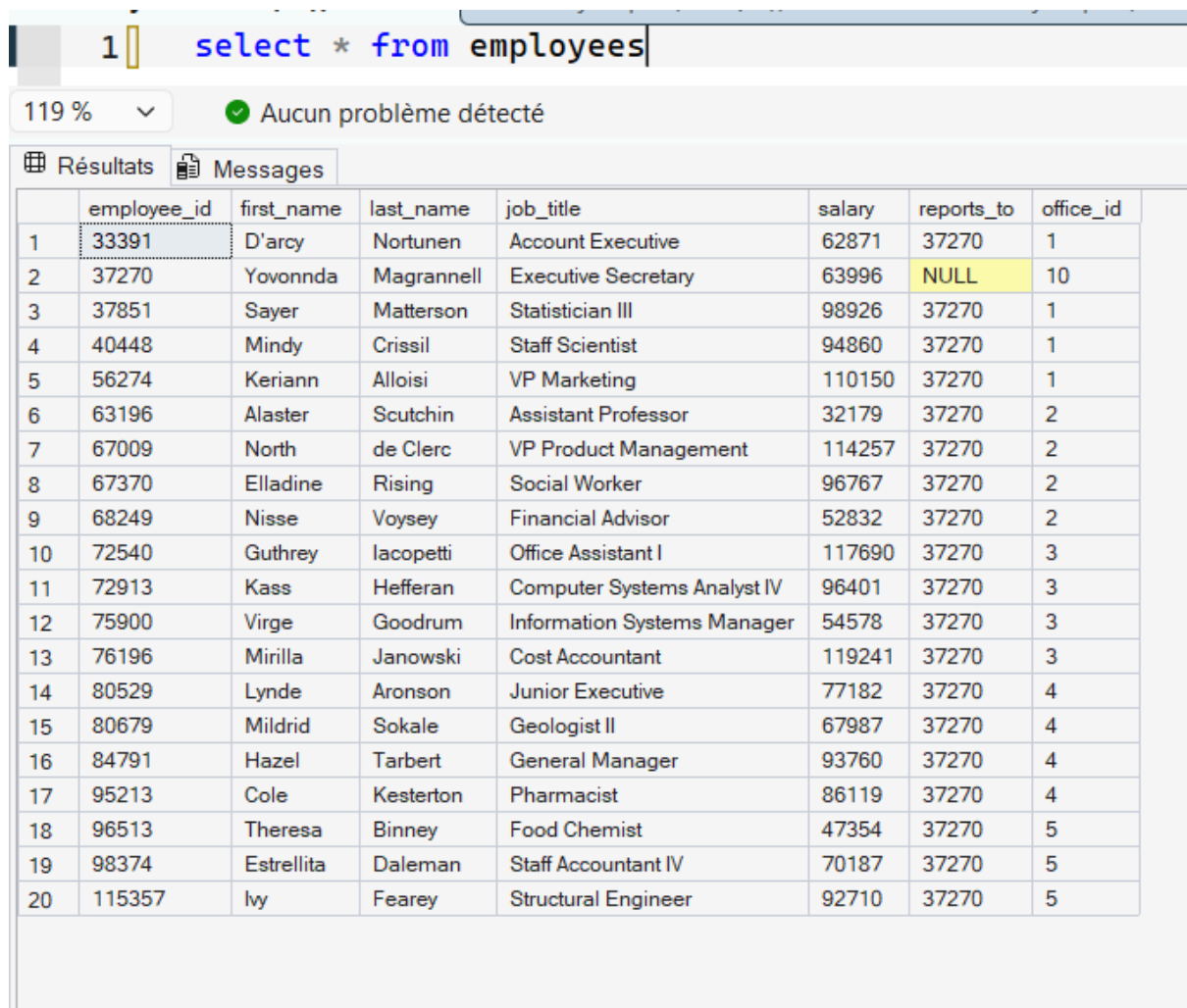


# Partie II : Questions SQL

## Niveau 1 (SELECT)

### Question 1

Afficher tous les employés.



The screenshot shows a SQL query editor with the query `select * from employees` entered. Below the query, a status bar indicates '119 %' zoom and 'Aucun problème détecté'. The results are displayed in a table with columns: employee\_id, first\_name, last\_name, job\_title, salary, reports\_to, and office\_id. The table contains 20 rows of employee data.

|    | employee_id | first_name | last_name  | job_title                   | salary | reports_to | office_id |
|----|-------------|------------|------------|-----------------------------|--------|------------|-----------|
| 1  | 33391       | D'arcy     | Nortunen   | Account Executive           | 62871  | 37270      | 1         |
| 2  | 37270       | Yovonnda   | Magrannell | Executive Secretary         | 63996  | NULL       | 10        |
| 3  | 37851       | Sayer      | Matterson  | Statistician III            | 98926  | 37270      | 1         |
| 4  | 40448       | Mindy      | Crissil    | Staff Scientist             | 94860  | 37270      | 1         |
| 5  | 56274       | Keriann    | Alloisi    | VP Marketing                | 110150 | 37270      | 1         |
| 6  | 63196       | Alaster    | Scutchin   | Assistant Professor         | 32179  | 37270      | 2         |
| 7  | 67009       | North      | de Clerc   | VP Product Management       | 114257 | 37270      | 2         |
| 8  | 67370       | Elladine   | Rising     | Social Worker               | 96767  | 37270      | 2         |
| 9  | 68249       | Nisse      | Voysey     | Financial Advisor           | 52832  | 37270      | 2         |
| 10 | 72540       | Guthrey    | Iacopetti  | Office Assistant I          | 117690 | 37270      | 3         |
| 11 | 72913       | Kass       | Hefferan   | Computer Systems Analyst IV | 96401  | 37270      | 3         |
| 12 | 75900       | Virge      | Goodrum    | Information Systems Manager | 54578  | 37270      | 3         |
| 13 | 76196       | Mirilla    | Janowski   | Cost Accountant             | 119241 | 37270      | 3         |
| 14 | 80529       | Lynde      | Aronson    | Junior Executive            | 77182  | 37270      | 4         |
| 15 | 80679       | Mildrid    | Sokale     | Geologist II                | 67987  | 37270      | 4         |
| 16 | 84791       | Hazel      | Tarbert    | General Manager             | 93760  | 37270      | 4         |
| 17 | 95213       | Cole       | Kesterton  | Pharmacist                  | 86119  | 37270      | 4         |
| 18 | 96513       | Theresa    | Binney     | Food Chemist                | 47354  | 37270      | 5         |
| 19 | 98374       | Estrellita | Daleman    | Staff Accountant IV         | 70187  | 37270      | 5         |
| 20 | 115357      | Ivy        | Fearey     | Structural Engineer         | 92710  | 37270      | 5         |

### Question 2

Afficher uniquement le prénom et le salaire des employés.

|           |                                          |        |
|-----------|------------------------------------------|--------|
| 1         | select First_name, Salary from employees |        |
| 19 %      | ✓ Aucun problème détecté                 |        |
| Résultats | Messages                                 |        |
|           | First_name                               | Salary |
| 1         | D'arcy                                   | 62871  |
| 2         | Yovonnda                                 | 63996  |
| 3         | Sayer                                    | 98926  |
| 4         | Mindy                                    | 94860  |
| 5         | Keriann                                  | 110150 |
| 6         | Alaster                                  | 32179  |
| 7         | North                                    | 114257 |
| 8         | Elladine                                 | 96767  |
| 9         | Nisse                                    | 52832  |
| 10        | Guthrey                                  | 117690 |
| 11        | Kass                                     | 96401  |
| 12        | Virge                                    | 54578  |
| 13        | Mirilla                                  | 119241 |
| 14        | Lynde                                    | 77182  |
| 15        | Mildrid                                  | 67987  |
| 16        | Hazel                                    | 93760  |
| 17        | Cole                                     | 86119  |
| 18        | Theresa                                  | 47354  |
| 19        | Estrellita                               | 70187  |
| 20        | Ivy                                      | 92710  |

### Question 3

Afficher tous les produits disponibles.

SQLQuery3.s... \Dell (65))\* SQLQuery2.sq... R\Dell (59))\* SQLQuery1.sq... R\Dell (78)

1 `select * from products where quantity_in_stock > 0;`

119 % ✓ Aucun problème détecté

Résultats Messages

|    | product_id | name                         | quantity_in_stock | unit_price |
|----|------------|------------------------------|-------------------|------------|
| 1  | 1          | Foam Dinner Plate            | 70                | 1.21       |
| 2  | 2          | Pork - Bacon,back Peameal    | 49                | 4.65       |
| 3  | 3          | Lettuce - Romaine, Heart     | 38                | 3.35       |
| 4  | 4          | Brocolinni - Gaylan, Chinese | 90                | 4.53       |
| 5  | 5          | Sauce - Ranch Dressing       | 94                | 1.63       |
| 6  | 6          | Petit Baguette               | 14                | 2.39       |
| 7  | 7          | Sweet Pea Sprouts            | 98                | 3.29       |
| 8  | 8          | Island Oasis - Raspberry     | 26                | 0.74       |
| 9  | 9          | Longan                       | 67                | 2.26       |
| 10 | 10         | Broom - Push                 | 6                 | 1.09       |

## Question 4

Afficher tous les clients.

SQLQuery3.s... \Dell (65))\* SQLQuery2.sq... R\Dell (59))\* SQLQuery1.sq... R\Dell (78))\* create-

1 `select * from customers;`

119 % ✓ Aucun problème détecté

Résultats Messages

|    | customer_id | first_name | last_name  | birth_date | phone        | address                | city             | state | points |
|----|-------------|------------|------------|------------|--------------|------------------------|------------------|-------|--------|
| 1  | 1           | Babara     | MacCaffrey | 1986-03-28 | 781-932-9754 | 0 Sage Terrace         | Waltham          | MA    | 2273   |
| 2  | 2           | Ines       | Brushfield | 1986-04-13 | 804-427-9456 | 14187 Commercial Trail | Hampton          | VA    | 947    |
| 3  | 3           | Freddi     | Boagey     | 1985-02-07 | 719-724-7869 | 251 Springs Junction   | Colorado Springs | CO    | 2967   |
| 4  | 4           | Ambur      | Roseburgh  | 1974-04-14 | 407-231-8017 | 30 Arapahoe Terrace    | Orlando          | FL    | 457    |
| 5  | 5           | Clemmie    | Betchley   | 1973-11-07 | NULL         | 5 Spohn Circle         | Arlington        | TX    | 3675   |
| 6  | 6           | Elka       | Twiddell   | 1991-09-04 | 312-480-8498 | 7 Manley Drive         | Chicago          | IL    | 3073   |
| 7  | 7           | Ilene      | Dowson     | 1964-08-30 | 615-641-4759 | 50 Lillian Crossing    | Nashville        | TN    | 1672   |
| 8  | 8           | Thacher    | Naseby     | 1993-07-17 | 941-527-3977 | 538 Mosinee Center     | Sarasota         | FL    | 205    |
| 9  | 9           | Romola     | Rumgay     | 1992-05-23 | 559-181-3744 | 3520 Ohio Trail        | Visalia          | CA    | 1486   |
| 10 | 10          | Levy       | Mynett     | 1969-10-13 | 404-246-3370 | 68 Lawn Avenue         | Atlanta          | GA    | 796    |

## Question 5

Afficher les factures.

| 1         | select * from invoices |             |           |               |               |              |            |              |
|-----------|------------------------|-------------|-----------|---------------|---------------|--------------|------------|--------------|
| 119 %     | Aucun problème détecté |             |           |               |               |              |            |              |
| Résultats | Messages               |             |           |               |               |              |            |              |
|           | invoice_id             | number      | client_id | invoice_total | payment_total | invoice_date | due_date   | payment_date |
| 1         | 1                      | 91-953-3396 | 2         | 101.79        | 0.00          | 2019-03-09   | 2019-03-29 | NULL         |
| 2         | 2                      | 03-898-6735 | 5         | 175.32        | 8.18          | 2019-06-11   | 2019-07-01 | 2019-02-12   |
| 3         | 3                      | 20-228-0335 | 5         | 147.99        | 0.00          | 2019-07-31   | 2019-08-20 | NULL         |
| 4         | 4                      | 56-934-0748 | 3         | 152.21        | 0.00          | 2019-03-08   | 2019-03-28 | NULL         |
| 5         | 5                      | 87-052-3121 | 5         | 169.36        | 0.00          | 2019-07-18   | 2019-08-07 | NULL         |
| 6         | 6                      | 75-587-6626 | 1         | 157.78        | 74.55         | 2019-01-29   | 2019-02-18 | 2019-01-03   |
| 7         | 7                      | 68-093-9863 | 3         | 133.87        | 0.00          | 2019-09-04   | 2019-09-24 | NULL         |
| 8         | 8                      | 78-145-1093 | 1         | 189.12        | 0.00          | 2019-05-20   | 2019-06-09 | NULL         |
| 9         | 9                      | 77-593-0081 | 5         | 172.17        | 0.00          | 2019-07-09   | 2019-07-29 | NULL         |
| 10        | 10                     | 48-266-1517 | 1         | 159.50        | 0.00          | 2019-06-30   | 2019-07-20 | NULL         |
| 11        | 11                     | 20-848-0181 | 3         | 126.15        | 0.03          | 2019-01-07   | 2019-01-27 | 2019-01-11   |
| 12        | 13                     | 41-666-1035 | 5         | 135.01        | 87.44         | 2019-06-25   | 2019-07-15 | 2019-01-26   |
| 13        | 15                     | 55-105-9605 | 3         | 167.29        | 80.31         | 2019-11-25   | 2019-12-15 | 2019-01-15   |
| 14        | 16                     | 10-451-8824 | 1         | 162.02        | 0.00          | 2019-03-30   | 2019-04-19 | NULL         |
| 15        | 17                     | 33-615-4694 | 3         | 126.38        | 68.10         | 2019-07-30   | 2019-08-19 | 2019-01-15   |
| 16        | 18                     | 52-269-9803 | 5         | 180.17        | 42.77         | 2019-05-23   | 2019-06-12 | 2019-01-08   |
| 17        | 19                     | 83-559-4105 | 1         | 134.47        | 0.00          | 2019-11-23   | 2019-12-13 | NULL         |

## Niveau 2 (WHERE)

### Question 6

Afficher les employés dont le salaire dépasse 90 000.

| 1         | select * from employees where salary > 90000; |            |           |                             |        |            |           |
|-----------|-----------------------------------------------|------------|-----------|-----------------------------|--------|------------|-----------|
| 119 %     | Aucun problème détecté                        |            |           |                             |        |            |           |
| Résultats | Messages                                      |            |           |                             |        |            |           |
|           | employee_id                                   | first_name | last_name | job_title                   | salary | reports_to | office_id |
| 1         | 37851                                         | Sayer      | Matterson | Statistician III            | 98926  | 37270      | 1         |
| 2         | 40448                                         | Mindy      | Crissil   | Staff Scientist             | 94860  | 37270      | 1         |
| 3         | 56274                                         | Keriann    | Alloisi   | VP Marketing                | 110150 | 37270      | 1         |
| 4         | 67009                                         | North      | de Clerc  | VP Product Management       | 114257 | 37270      | 2         |
| 5         | 67370                                         | Elladine   | Rising    | Social Worker               | 96767  | 37270      | 2         |
| 6         | 72540                                         | Guthrey    | Iacopetti | Office Assistant I          | 117690 | 37270      | 3         |
| 7         | 72913                                         | Kass       | Hefferan  | Computer Systems Analyst IV | 96401  | 37270      | 3         |
| 8         | 76196                                         | Mirilla    | Janowski  | Cost Accountant             | 119241 | 37270      | 3         |
| 9         | 84791                                         | Hazel      | Tarbert   | General Manager             | 93760  | 37270      | 4         |
| 10        | 115357                                        | Ivy        | Fearey    | Structural Engineer         | 92710  | 37270      | 5         |

---

## Question 7

Afficher les employés travaillant dans le bureau numéro 3.

```
1 | select * from employees where office_id = 3;
```

19 % Aucun problème détecté

Résultats Messages

|  | employee_id | first_name | last_name | job_title                   | salary | reports_to | office_id |
|--|-------------|------------|-----------|-----------------------------|--------|------------|-----------|
|  | 72540       | Guthrey    | Iacopetti | Office Assistant I          | 117690 | 37270      | 3         |
|  | 72913       | Kass       | Hefferan  | Computer Systems Analyst IV | 96401  | 37270      | 3         |
|  | 75900       | Virge      | Goodrum   | Information Systems Manager | 54578  | 37270      | 3         |
|  | 76196       | Mirilla    | Janowski  | Cost Accountant             | 119241 | 37270      | 3         |

---

## Question 8

Afficher les produits dont le prix est supérieur à 3 dollars.

```
1 | select * from employees where office_id = 3;
```

19 % Aucun problème détecté

Résultats Messages

|  | employee_id | first_name | last_name | job_title                   | salary | reports_to | office_id |
|--|-------------|------------|-----------|-----------------------------|--------|------------|-----------|
|  | 72540       | Guthrey    | Iacopetti | Office Assistant I          | 117690 | 37270      | 3         |
|  | 72913       | Kass       | Hefferan  | Computer Systems Analyst IV | 96401  | 37270      | 3         |
|  | 75900       | Virge      | Goodrum   | Information Systems Manager | 54578  | 37270      | 3         |
|  | 76196       | Mirilla    | Janowski  | Cost Accountant             | 119241 | 37270      | 3         |

---

## Question 9

Afficher les clients habitant dans l'État CA.

1

```
select * from customers where state = 'CA';
```

119 %

Aucun problème détecté

Résultats

Messages

|   | customer_id | first_name | last_name | birth_date | phone        | address         | city    | state | points |
|---|-------------|------------|-----------|------------|--------------|-----------------|---------|-------|--------|
| 1 | 9           | Romola     | Rumgay    | 1992-05-23 | 559-181-3744 | 3520 Ohio Trail | Visalia | CA    | 1486   |

## Question 10

Afficher les commandes passées après le 1 janvier 2018.

SQLQuery3.s... \Dell (65))\* SQLQuery2.sq...R\Dell (59))\* SQLQuery1.sq...R\Dell (78))\*

1 select \* from orders where order\_date > '2018-01-01'

119 % Aucun problème détecté

Résultats Messages

|   | order_id | customer_id | order_date | status | comments                                           | shipped_date | shipper_id |
|---|----------|-------------|------------|--------|----------------------------------------------------|--------------|------------|
| 1 | 1        | 6           | 2019-01-30 | 1      | NULL                                               | NULL         | NULL       |
| 2 | 2        | 7           | 2018-08-02 | 2      | NULL                                               | 2018-08-03   | 4          |
| 3 | 6        | 10          | 2018-11-18 | 1      | Aliquam erat volutpat. In congue.                  | NULL         | NULL       |
| 4 | 7        | 2           | 2018-09-22 | 2      | NULL                                               | 2018-09-23   | 4          |
| 5 | 8        | 5           | 2018-06-08 | 1      | Mauris enim leo, rhoncus sed, vestibulum sit am... | NULL         | NULL       |
| 6 | 10       | 6           | 2018-04-22 | 2      | NULL                                               | 2018-04-23   | 2          |

## Niveau 3 (ORDER BY)

### Question 11

Afficher les employés du plus grand salaire au plus petit.

SQLQuery3.s... (Def (65)) x SQLQuery2.sq...K(Def (59)) SQLQuery1.sq...K(Def (70))

1 | select \* from employees order by salary desc;

119 % Aucun problème détecté

Résultats Messages

|    | employee_id | first_name | last_name  | job_title                   | salary | reports_to | office_id |
|----|-------------|------------|------------|-----------------------------|--------|------------|-----------|
| 1  | 76196       | Mirilla    | Janowski   | Cost Accountant             | 119241 | 37270      | 3         |
| 2  | 72540       | Guthrey    | Iacopetti  | Office Assistant I          | 117690 | 37270      | 3         |
| 3  | 67009       | North      | de Clerc   | VP Product Management       | 114257 | 37270      | 2         |
| 4  | 56274       | Keriann    | Alloisi    | VP Marketing                | 110150 | 37270      | 1         |
| 5  | 37851       | Sayer      | Matterson  | Statistician III            | 98926  | 37270      | 1         |
| 6  | 67370       | Elladine   | Rising     | Social Worker               | 96767  | 37270      | 2         |
| 7  | 72913       | Kass       | Hefferan   | Computer Systems Analyst IV | 96401  | 37270      | 3         |
| 8  | 40448       | Mindy      | Crissil    | Staff Scientist             | 94860  | 37270      | 1         |
| 9  | 84791       | Hazel      | Tarbert    | General Manager             | 93760  | 37270      | 4         |
| 10 | 115357      | Ivy        | Fearey     | Structural Engineer         | 92710  | 37270      | 5         |
| 11 | 95213       | Cole       | Kesterton  | Pharmacist                  | 86119  | 37270      | 4         |
| 12 | 80529       | Lynde      | Aronson    | Junior Executive            | 77182  | 37270      | 4         |
| 13 | 98374       | Estrellita | Daleman    | Staff Accountant IV         | 70187  | 37270      | 5         |
| 14 | 80679       | Mildrid    | Sokale     | Geologist II                | 67987  | 37270      | 4         |
| 15 | 37270       | Yovonnda   | Magrannell | Executive Secretary         | 63996  | NULL       | 10        |
| 16 | 33391       | D'arcy     | Nortunen   | Account Executive           | 62871  | 37270      | 1         |
| 17 | 75900       | Virge      | Goodrum    | Information Systems Manager | 54578  | 37270      | 3         |
| 18 | 68249       | Nisse      | Voysey     | Financial Advisor           | 52832  | 37270      | 2         |
| 19 | 96513       | Theresa    | Binney     | Food Chemist                | 47354  | 37270      | 5         |
| 20 | 63196       | Alaster    | Scutchin   | Assistant Professor         | 32179  | 37270      | 2         |

## Question 12

Afficher les produits classés par prix croissant.

SQLQuery3.s... \Dell (65))\* X SQLQuery2.sq...R\Dell (59))\* SQLQuery1.sq...R\De

1 `select * from products order by unit_price asc;`

119 % ✓ Aucun problème détecté

Résultats Messages

|    | product_id | name                         | quantity_in_stock | unit_price |
|----|------------|------------------------------|-------------------|------------|
| 1  | 8          | Island Oasis - Raspberry     | 26                | 0.74       |
| 2  | 10         | Broom - Push                 | 6                 | 1.09       |
| 3  | 1          | Foam Dinner Plate            | 70                | 1.21       |
| 4  | 5          | Sauce - Ranch Dressing       | 94                | 1.63       |
| 5  | 9          | Longan                       | 67                | 2.26       |
| 6  | 6          | Petit Baguette               | 14                | 2.39       |
| 7  | 7          | Sweet Pea Sprouts            | 98                | 3.29       |
| 8  | 3          | Lettuce - Romaine, Heart     | 38                | 3.35       |
| 9  | 4          | Brocolinni - Gaylan, Chinese | 90                | 4.53       |
| 10 | 2          | Pork - Bacon,back Peameal    | 49                | 4.65       |

## Question 13

Afficher les clients classés par nombre de points décroissant.

1 `select * from customers order by points desc;`

119 % ✓ Aucun problème détecté

Résultats Messages

|    | customer_id | first_name | last_name  | birth_date | phone        | address                | city             | state | points |
|----|-------------|------------|------------|------------|--------------|------------------------|------------------|-------|--------|
| 1  | 5           | Clemmie    | Betchley   | 1973-11-07 | NULL         | 5 Spohn Circle         | Arlington        | TX    | 3675   |
| 2  | 6           | Elka       | Twiddell   | 1991-09-04 | 312-480-8498 | 7 Manley Drive         | Chicago          | IL    | 3073   |
| 3  | 3           | Freddi     | Boagey     | 1985-02-07 | 719-724-7869 | 251 Springs Junction   | Colorado Springs | CO    | 2967   |
| 4  | 1           | Babara     | MacCaffrey | 1986-03-28 | 781-932-9754 | 0 Sage Terrace         | Waltham          | MA    | 2273   |
| 5  | 7           | Ilene      | Dowson     | 1964-08-30 | 615-641-4759 | 50 Lillian Crossing    | Nashville        | TN    | 1672   |
| 6  | 9           | Romola     | Rumgay     | 1992-05-23 | 559-181-3744 | 3520 Ohio Trail        | Visalia          | CA    | 1486   |
| 7  | 2           | Ines       | Brushfield | 1986-04-13 | 804-427-9456 | 14187 Commercial Trail | Hampton          | VA    | 947    |
| 8  | 10          | Levy       | Mynett     | 1969-10-13 | 404-246-3370 | 68 Lawn Avenue         | Atlanta          | GA    | 796    |
| 9  | 4           | Ambur      | Roseburgh  | 1974-04-14 | 407-231-8017 | 30 Arapahoe Terrace    | Orlando          | FL    | 457    |
| 10 | 8           | Thacher    | Naseby     | 1993-07-17 | 941-527-3977 | 538 Mosinee Center     | Sarasota         | FL    | 205    |

## Niveau 4 (LIMIT)



## Question 14

Afficher les 5 employés les mieux payés.

```
1 | select * from employees order by salary desc OFFSET 0 ROWS FETCH NEXT 5 ROWS ONLY
```

119 % Aucun problème détecté

Résultats Messages

|   | employee_id | first_name | last_name | job_title             | salary | reports_to | office_id |
|---|-------------|------------|-----------|-----------------------|--------|------------|-----------|
| 1 | 76196       | Mirilla    | Janowski  | Cost Accountant       | 119241 | 37270      | 3         |
| 2 | 72540       | Guthrey    | Iacopetti | Office Assistant I    | 117690 | 37270      | 3         |
| 3 | 67009       | North      | de Clerc  | VP Product Management | 114257 | 37270      | 2         |
| 4 | 56274       | Kerian     | Alloisi   | VP Marketing          | 110150 | 37270      | 1         |
| 5 | 37851       | Sayer      | Matterson | Statistician III      | 98926  | 37270      | 1         |

## Question 15

Afficher les 3 produits les moins chers.

```
1 | select * from products order by unit_price asc OFFSET 0 ROWS FETCH NEXT 3 ROWS ONLY
```

119 % Aucun problème détecté

Résultats Messages

|   | product_id | name                     | quantity_in_stock | unit_price |
|---|------------|--------------------------|-------------------|------------|
| 1 | 8          | Island Oasis - Raspberry | 26                | 0.74       |
| 2 | 10         | Broom - Push             | 6                 | 1.09       |
| 3 | 1          | Foam Dinner Plate        | 70                | 1.21       |

## Niveau 5 (Fonctions d'agrégation)

### Question 16

Quel est le salaire moyen des employés ?

```
1 | select avg(salary) as 'salaire moyen' from employees
```

119 % Aucun problème détecté

Résultats Messages

|   | salaire moyen |
|---|---------------|
| 1 | 82502         |

## Question 17

Quel est le salaire maximal ?

|           |                                                                     |
|-----------|---------------------------------------------------------------------|
| 1         | <code>select max(salary) as 'salaire maximal' from employees</code> |
| 119 %     | Aucun problème détecté                                              |
| Résultats | Messages                                                            |
|           | salaire maximal                                                     |
| 1         | 119241                                                              |

## Question 18

Quel est le salaire minimal ?

|           |                                                                     |
|-----------|---------------------------------------------------------------------|
| 1         | <code>select min(salary) as 'salaire minimal' from employees</code> |
| 119 %     | Aucun problème détecté                                              |
| Résultats | Messages                                                            |
|           | salaire minimal                                                     |
| 1         | 32179                                                               |

## Question 19

Combien y a-t-il d'employés ?

|           |                                                                                      |
|-----------|--------------------------------------------------------------------------------------|
| 1         | <code>select count(employee_id) as 'Nombre total des employés' from employees</code> |
| 119 %     | Aucun problème détecté                                                               |
| Résultats | Messages                                                                             |
|           | Nombre total des employés                                                            |
| 1         | 20                                                                                   |

## Question 20

Quelle est la valeur totale du stock ?

(quantité × prix)

|   |                                                                                                      |
|---|------------------------------------------------------------------------------------------------------|
| 1 | <code>select sum(quantity_in_stock * unit_price) as "La valeur totale du stock" from products</code> |
| 2 |                                                                                                      |
| 3 |                                                                                                      |

119 % ✓ Aucun problème détecté

| Résultats                                                                                                   | Messages                  |                           |   |         |  |
|-------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|---|---------|--|
| <table> <tr> <th></th><th>La valeur totale du stock</th></tr> <tr> <td>1</td><td>1533.85</td></tr> </table> |                           | La valeur totale du stock | 1 | 1533.85 |  |
|                                                                                                             | La valeur totale du stock |                           |   |         |  |
| 1                                                                                                           | 1533.85                   |                           |   |         |  |

## Niveau 6 (GROUP BY)

### Question 21

Calculer le salaire moyen par bureau.

1

```
select avg(salary) as "Salaire moyen", office_id as "Bureau" from employees group by office_id
```

119 %

Aucun problème détecté

Résultats

Messages

|   | Salaire moyen | Bureau |
|---|---------------|--------|
| 1 | 91701         | 1      |
| 2 | 74008         | 2      |
| 3 | 96977         | 3      |
| 4 | 81262         | 4      |
| 5 | 70083         | 5      |
| 6 | 63996         | 10     |

### Question 22

Compter le nombre d'employés par bureau.

1

```
select count(employee_id) as "Nombre d'employés", office_id as "Bureau" from employees group by office_id
```

119 %

Aucun problème détecté

Lig. : 1, Car. : 48

Résultats

Messages

|   | Nombre d'employés | Bureau |
|---|-------------------|--------|
| 1 | 4                 | 1      |
| 2 | 4                 | 2      |
| 3 | 4                 | 3      |
| 4 | 4                 | 4      |
| 5 | 3                 | 5      |
| 6 | 1                 | 10     |

### Question 23

Compter le nombre de clients par État.

1

select count(customer\_id) as "Nombre de clients", state as "Etat" from customers group by state

119 %

Aucun problème détecté

Lig

Résultats

Messages

|   | Nombre de clients | Etat |
|---|-------------------|------|
| 1 | 1                 | CA   |
| 2 | 1                 | CO   |
| 3 | 2                 | FL   |
| 4 | 1                 | GA   |
| 5 | 1                 | IL   |
| 6 | 1                 | MA   |
| 7 | 1                 | TN   |
| 8 | 1                 | TX   |
| 9 | 1                 | VA   |

## Question 24

Calculer le montant total facturé par client.

create-db-sto...GR\Dell (69))

SQLQuery1.s...\Dell (94))\*

```
1 select sum(quantity * unit_price) as "Montant total", customer_id as Clients from orders
2 inner join order_items on orders.order_id = order_items.order_id group by customer_id
```

119 %

Aucun problème détecté

Résultats Messages

|   | Montant total | Clients |
|---|---------------|---------|
| 1 | 157.92        | 2       |
| 2 | 60.73         | 5       |
| 3 | 113.58        | 6       |
| 4 | 30.72         | 7       |
| 5 | 91.20         | 8       |
| 6 | 117.41        | 10      |

## Question 25

Calculer le nombre de commandes par client.

```
1 select count(distinct order_id) as "Nombre de commande", customer_id as Clients
2 from orders group by customer_id
```

119 % Aucun problème détecté

Résultats Messages

|   |   |    |
|---|---|----|
| 6 | 2 | 10 |
|---|---|----|

|   | Montant total | Clients |
|---|---------------|---------|
| 1 | 157.92        | 2       |
| 2 | 60.73         | 5       |
| 3 | 113.58        | 6       |
| 4 | 30.72         | 7       |
| 5 | 91.20         | 8       |
| 6 | 117.41        | 10      |

## Niveau 7 (HAVING)

## Question 26

Afficher les bureaux possédant plus de deux employés.

|   |                                                                                      |  |
|---|--------------------------------------------------------------------------------------|--|
| 1 | select count(employee_id) as "Nombre d'employés", office_id as Bureau from employees |  |
| 2 | group by office_id having count(employee_id) > 2                                     |  |

119 % ● Aucun problème détecté

|   | Nombre d'employés | Bureau |
|---|-------------------|--------|
| 1 | 4                 | 1      |
| 2 | 4                 | 2      |
| 3 | 4                 | 3      |
| 4 | 4                 | 4      |
| 5 | 3                 | 5      |

## Question 27

Afficher les clients ayant plus d'une facture.

|   |                                                                                    |  |
|---|------------------------------------------------------------------------------------|--|
| 1 | select count(order_id) as "Nombre de factures", customer_id as clients from orders |  |
| 2 | group by customer_id having count(order_id) >1                                     |  |

119 % ● Aucun problème détecté

|   | Nombre de factures | clients |
|---|--------------------|---------|
| 1 | 2                  | 2       |
| 2 | 2                  | 5       |
| 3 | 2                  | 6       |
| 4 | 2                  | 10      |

## Question 28

Afficher les États ayant au moins deux clients.

|   |                                                                                 |  |
|---|---------------------------------------------------------------------------------|--|
| 1 | select count(customer_id) as "Nombre de clients", state as Etats from customers |  |
| 2 | group by state having count(customer_id) >= 2;                                  |  |

119 % ● Aucun problème détecté

|   | Nombre de clients | Etats |
|---|-------------------|-------|
| 1 | 2                 | FL    |

## Niveau 8 (Jointures)

### Question 29

Afficher :

- prénom
- nom
- ville du bureau

|   |                                                                |
|---|----------------------------------------------------------------|
| 1 | SELECT first_name, last_name, city FROM employees              |
| 2 | inner join offices on employees.office_id = offices.office_id; |

119 % ✓ Aucun problème détecté

|    | first_name | last_name  | city          |
|----|------------|------------|---------------|
| 1  | D'arcy     | Nortunen   | Cincinnati    |
| 2  | Yovonnda   | Magrannell | Savannah      |
| 3  | Sayer      | Matterson  | Cincinnati    |
| 4  | Mindy      | Crissil    | Cincinnati    |
| 5  | Keriann    | Alloisi    | Cincinnati    |
| 6  | Alaster    | Scutchin   | New York City |
| 7  | North      | de Clerc   | New York City |
| 8  | Elladine   | Rising     | New York City |
| 9  | Nisse      | Voysey     | New York City |
| 10 | Guthrey    | Iacopetti  | Richmond      |
| 11 | Kass       | Hefferan   | Richmond      |
| 12 | Virge      | Goodrum    | Richmond      |
| 13 | Mirilla    | Janowski   | Richmond      |
| 14 | Lynde      | Aronson    | Cincinnati    |
| 15 | Mildrid    | Sokale     | Cincinnati    |
| 16 | Hazel      | Tarbert    | Cincinnati    |
| 17 | Cole       | Kesterton  | Cincinnati    |

## Question 30

Afficher :

- nom du client
- numéro de commande
- date de commande

|   |                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------|
| 1 | select Concat(First_name, ' ',last_name) as "Nom complet", order_id as "Numéro de commande",            |
| 2 | order_date as "Date de commande" from customers c inner join orders o on c.customer_id = o.customer_id; |

119 % ✓ Aucun problème détecté Lig.: 2, Car.: 104 | 1

|   | Nom complet      | Numéro de commande | Date de commande |
|---|------------------|--------------------|------------------|
| 1 | Elka Twiddell    | 1                  | 2019-01-30       |
| 2 | Ilene Dowson     | 2                  | 2018-08-02       |
| 3 | Thacher Naseby   | 3                  | 2017-12-01       |
| 4 | Ines Brushfield  | 4                  | 2017-01-22       |
| 5 | Clemmie Betchley | 5                  | 2017-08-25       |
| 6 | Levy Mynett      | 6                  | 2018-11-18       |
| 7 | Ines Brushfield  | 7                  | 2018-09-22       |
| 8 | Clemmie Betchley | 8                  | 2018-06-08       |

## Question 31

Afficher :

- numéro de commande
- nom du produit
- quantité

|   |                                                                                            |
|---|--------------------------------------------------------------------------------------------|
| 1 | select o.order_id as "Numéro de commande", p.name as "Nom produit", o.quantity as Quantité |
| 2 | from products p inner join order_items o on p.product_id = o.product_id                    |

119 %    ● Aucun problème détecté

|   | Numéro de commande | Nom produit                  | Quantité |
|---|--------------------|------------------------------|----------|
| 1 | 1                  | Brocolinni - Gaylan, Chinese | 4        |
| 2 | 2                  | Foam Dinner Plate            | 2        |
| 3 | 2                  | Brocolinni - Gaylan, Chinese | 4        |
| 4 | 2                  | Petit Baguette               | 2        |
| 5 | 3                  | Lettuce - Romaine, Heart     | 10       |
| 6 | 4                  | Lettuce - Romaine, Heart     | 7        |
| 7 | 4                  | Broom - Push                 | 7        |
| 8 | 5                  | Pork - Bacon,back Peameal    | 3        |

## Question 32

Afficher :

- numéro de facture
- nom du client
- montant de la facture

|   |                                                                                            |
|---|--------------------------------------------------------------------------------------------|
| 1 | select o.order_id as "Numéron de facture", c.first_name as "Nom client",                   |
| 2 | sum(quantity * unit_price) as "Montant de la Facture" from customers c inner join orders o |
| 3 | on c.customer_id = o.customer_id inner join order_items oi on o.order_id = oi.order_id     |
| 4 | group by o.order_id, c.first_name                                                          |

119 %    ● Aucun problème détecté

|   | Numéron de facture | Nom client | Montant de la Facture |
|---|--------------------|------------|-----------------------|
| 1 | 1                  | Elka       | 14.96                 |
| 2 | 2                  | Ilene      | 30.72                 |
| 3 | 3                  | Thacher    | 91.20                 |
| 4 | 4                  | Ines       | 93.73                 |
| 5 | 5                  | Clemmie    | 29.67                 |
| 6 | 6                  | Levy       | 81.01                 |
| 7 | 7                  | Ines       | 64.19                 |
| 8 | 8                  | Clemmie    | 31.06                 |

## Question 33

Afficher :

- paiement
- méthode de paiement
- nom du client

|   |                                                                                                |
|---|------------------------------------------------------------------------------------------------|
| 1 | select p.payment_id as Paiement, c.name as "Nom client", pm.name as "Méthode de paiement" from |
| 2 | clients c inner join payments p on c.client_id = p.client_id                                   |
| 3 | inner join payment_methods pm on p.payment_method = pm.payment_method_id;                      |

119 %    ● Aucun problème détecté    Lig.: 3, Car.: 74

|   | Paiement | Nom client  | Paiement Methode |
|---|----------|-------------|------------------|
| 1 | 1        | Topiclounge | Credit Card      |
| 2 | 2        | Vinte       | Credit Card      |
| 3 | 3        | Yadel       | Credit Card      |
| 4 | 4        | Topiclounge | Credit Card      |
| 5 | 5        | Yadel       | Credit Card      |
| 6 | 6        | Yadel       | Credit Card      |
| 7 | 7        | Topiclounge | Credit Card      |
| 8 | 8        | Topiclounge | Cash             |

### Question 34

Afficher le nom du manager de chaque employé.

## Niveau 9 (LEFT JOIN)

### Question 35

Afficher tous les clients, même ceux qui n'ont jamais passé de commande.

1

Select \* from customers c left join orders o

2

on c.customer\_id = o.customer\_id

119 %

Aucun problème détecté

Lig. : 2, Car. : 33

TABS

CRLF

UTF-8

Résultats

Messages

|    | customer_id | first_name | last_name  | birth_date | phone        | address                | city             | state | points | order_id | customer_id | order_date | status | comments                                           | shipped_date |
|----|-------------|------------|------------|------------|--------------|------------------------|------------------|-------|--------|----------|-------------|------------|--------|----------------------------------------------------|--------------|
| 1  | 1           | Babara     | MacCaffrey | 1986-03-28 | 781-932-9754 | 0 Sage Terrace         | Waltham          | MA    | 2273   | NULL     | NULL        | NULL       | NULL   | NULL                                               | NULL         |
| 2  | 2           | Ines       | Brushfield | 1986-04-13 | 804-427-9456 | 14187 Commercial Trail | Hampton          | VA    | 947    | 4        | 2           | 2017-01-22 | 1      | NULL                                               | NULL         |
| 3  | 2           | Ines       | Brushfield | 1986-04-13 | 804-427-9456 | 14187 Commercial Trail | Hampton          | VA    | 947    | 7        | 2           | 2018-09-22 | 2      | NULL                                               | 2018-09-23   |
| 4  | 3           | Freddi     | Boagey     | 1985-02-07 | 719-724-7869 | 251 Springs Junction   | Colorado Springs | CO    | 2967   | NULL     | NULL        | NULL       | NULL   | NULL                                               | NULL         |
| 5  | 4           | Ambur      | Roseburgh  | 1974-04-14 | 407-231-8017 | 30 Arapahoe Terrace    | Orlando          | FL    | 457    | NULL     | NULL        | NULL       | NULL   | NULL                                               | NULL         |
| 6  | 5           | Clemmie    | Betchley   | 1973-11-07 | NULL         | 5 Spohn Circle         | Arlington        | TX    | 3675   | 5        | 5           | 2017-08-25 | 2      |                                                    | 2017-08-26   |
| 7  | 5           | Clemmie    | Betchley   | 1973-11-07 | NULL         | 5 Spohn Circle         | Arlington        | TX    | 3675   | 8        | 5           | 2018-06-08 | 1      | Mauris enim leo. rhoncus sed, vestibulum sit am... | NULL         |
| 8  | 6           | Elka       | Twiddell   | 1991-09-04 | 312-480-8498 | 7 Manley Drive         | Chicago          | IL    | 3073   | 1        | 6           | 2019-01-30 | 1      | NULL                                               | NULL         |
| 9  | 6           | Elka       | Twiddell   | 1991-09-04 | 312-480-8498 | 7 Manley Drive         | Chicago          | IL    | 3073   | 10       | 6           | 2018-04-22 | 2      | NULL                                               | 2018-04-23   |
| 10 | 7           | Ilene      | Dowson     | 1964-08-30 | 615-641-4759 | 50 Lillian Crossing    | Nashville        | TN    | 1672   | 2        | 7           | 2018-08-02 | 2      | NULL                                               | 2018-08-03   |
| 11 | 8           | Thacher    | Naseby     | 1993-07-17 | 941-527-3977 | 538 Mosinee Center     | Sarasota         | FL    | 205    | 3        | 8           | 2017-12-01 | 1      | NULL                                               | NULL         |
| 12 | 9           | Romola     | Rumgay     | 1992-05-23 | 559-181-3744 | 3520 Ohio Trail        | Visalia          | CA    | 1486   | NULL     | NULL        | NULL       | NULL   | NULL                                               | NULL         |
| 13 | 10          | Lewy       | Mynett     | 1969-10-13 | 404-246-3370 | 68 Lawn Avenue         | Atlanta          | GA    | 796    | 6        | 10          | 2018-11-18 | 1      | Aliquam erat volutpat. In congue.                  | NULL         |
| 14 | 10          | Lewy       | Mynett     | 1969-10-13 | 404-246-3370 | 68 Lawn Avenue         | Atlanta          | GA    | 796    | 9        | 10          | 2017-07-05 | 2      | Nulla mollis molestie lorem. Quisque ut erat.      | 2017-07-06   |

### Question 36

Afficher tous les produits, même ceux jamais commandés.



```

1 | Select * from products p left join order_items oi
2 | on p.product_id = oi.product_id;

```

119 % Aucun problème détecté

Résultats Messages

|    | product_id | name                         | quantity_in_stock | unit_price | order_id | product_id | quantity | unit_price |
|----|------------|------------------------------|-------------------|------------|----------|------------|----------|------------|
| 1  | 1          | Foam Dinner Plate            | 70                | 1.21       | 2        | 1          | 2        | 9.10       |
| 2  | 1          | Foam Dinner Plate            | 70                | 1.21       | 6        | 1          | 4        | 8.65       |
| 3  | 1          | Foam Dinner Plate            | 70                | 1.21       | 10       | 1          | 10       | 6.01       |
| 4  | 2          | Pork - Bacon,back Peameal    | 49                | 4.65       | 5        | 2          | 3        | 9.89       |
| 5  | 2          | Pork - Bacon,back Peameal    | 49                | 4.65       | 6        | 2          | 4        | 3.28       |
| 6  | 3          | Lettuce - Romaine, Heart     | 38                | 3.35       | 3        | 3          | 10       | 9.12       |
| 7  | 3          | Lettuce - Romaine, Heart     | 38                | 3.35       | 4        | 3          | 7        | 6.99       |
| 8  | 3          | Lettuce - Romaine, Heart     | 38                | 3.35       | 6        | 3          | 4        | 7.46       |
| 9  | 3          | Lettuce - Romaine, Heart     | 38                | 3.35       | 7        | 3          | 7        | 9.17       |
| 10 | 4          | Brocolinni - Gaylan, Chinese | 90                | 4.53       | 1        | 4          | 4        | 3.74       |
| 11 | 4          | Brocolinni - Gaylan, Chinese | 90                | 4.53       | 2        | 4          | 4        | 1.66       |
| 12 | 5          | Sauce - Ranch Dressing       | 94                | 1.63       | 6        | 5          | 1        | 3.45       |
| 13 | 5          | Sauce - Ranch Dressing       | 94                | 1.63       | 8        | 5          | 2        | 6.94       |
| 14 | 6          | Petit Baguette               | 14                | 2.39       | 2        | 6          | 2        | 2.94       |
| 15 | 6          | Petit Baguette               | 14                | 2.39       | 9        | 6          | 5        | 7.28       |
| 16 | 7          | Sweet Pea Sprouts            | 98                | 3.29       | NULL     | NULL       | NULL     | NULL       |
| 17 | 8          | Island Oasis - Raspberry     | 26                | 0.74       | 8        | 8          | 2        | 8.59       |
| 18 | 9          | Longan                       | 67                | 2.26       | 10       | 9          | 9        | 4.28       |
| 19 | 10         | Broom - Push                 | 6                 | 1.09       | 4        | 10         | 7        | 6.40       |

## Question 37

Afficher tous les bureaux, même sans employé.

```

1 | SELECT * from offices o left join employees e on o.office_id = e.office_id

```

119 % Aucun problème détecté

Résultats Messages

|    | office_id | address             | city          | state | employee_id | first_name | last_name  | job_title                   | salary | reports_to | office_id |
|----|-----------|---------------------|---------------|-------|-------------|------------|------------|-----------------------------|--------|------------|-----------|
| 1  | 1         | 03 Reinke Trail     | Cincinnati    | OH    | 33391       | D'arcy     | Nortunen   | Account Executive           | 62871  | 37270      | 1         |
| 2  | 1         | 03 Reinke Trail     | Cincinnati    | OH    | 37851       | Sayer      | Matterson  | Statistician III            | 98926  | 37270      | 1         |
| 3  | 1         | 03 Reinke Trail     | Cincinnati    | OH    | 40448       | Mindy      | Crissil    | Staff Scientist             | 94860  | 37270      | 1         |
| 4  | 1         | 03 Reinke Trail     | Cincinnati    | OH    | 56274       | Keriann    | Alloisi    | VP Marketing                | 110150 | 37270      | 1         |
| 5  | 2         | 5507 Becker Terrace | New York City | NY    | 63196       | Alaster    | Scutchin   | Assistant Professor         | 32179  | 37270      | 2         |
| 6  | 2         | 5507 Becker Terrace | New York City | NY    | 67009       | North      | de Clerc   | VP Product Management       | 114257 | 37270      | 2         |
| 7  | 2         | 5507 Becker Terrace | New York City | NY    | 67370       | Elladine   | Rising     | Social Worker               | 96767  | 37270      | 2         |
| 8  | 2         | 5507 Becker Terrace | New York City | NY    | 68249       | Nisse      | Voysey     | Financial Advisor           | 52832  | 37270      | 2         |
| 9  | 3         | 54 Northland Court  | Richmond      | VA    | 72540       | Guthrey    | Iacopetti  | Office Assistant I          | 117690 | 37270      | 3         |
| 10 | 3         | 54 Northland Court  | Richmond      | VA    | 72913       | Kass       | Hefferan   | Computer Systems Analyst IV | 96401  | 37270      | 3         |
| 11 | 3         | 54 Northland Court  | Richmond      | VA    | 75900       | Virge      | Goodrum    | Information Systems Manager | 54578  | 37270      | 3         |
| 12 | 3         | 54 Northland Court  | Richmond      | VA    | 76196       | Mirilla    | Janowski   | Cost Accountant             | 119241 | 37270      | 3         |
| 13 | 4         | 08 South Crossing   | Cincinnati    | OH    | 80529       | Lynde      | Aronson    | Junior Executive            | 77182  | 37270      | 4         |
| 14 | 4         | 08 South Crossing   | Cincinnati    | OH    | 80679       | Mildrid    | Sokale     | Geologist II                | 67987  | 37270      | 4         |
| 15 | 4         | 08 South Crossing   | Cincinnati    | OH    | 84791       | Hazel      | Tarbert    | General Manager             | 93760  | 37270      | 4         |
| 16 | 4         | 08 South Crossing   | Cincinnati    | OH    | 95213       | Cole       | Kesterton  | Pharmacist                  | 86119  | 37270      | 4         |
| 17 | 5         | 553 Maple Drive     | Minneapolis   | MN    | 96513       | Theresa    | Binney     | Food Chemist                | 47354  | 37270      | 5         |
| 18 | 5         | 553 Maple Drive     | Minneapolis   | MN    | 98374       | Estrellita | Daleman    | Staff Accountant IV         | 70187  | 37270      | 5         |
| 19 | 5         | 553 Maple Drive     | Minneapolis   | MN    | 115357      | Ivy        | Fearey     | Structural Engineer         | 92710  | 37270      | 5         |
| 20 | 6         | 23 North Plaza      | Aurora        | CO    | NULL        | NULL       | NULL       | NULL                        | NULL   | NULL       | NULL      |
| 21 | 7         | 9658 Wayridge Court | Boise         | ID    | NULL        | NULL       | NULL       | NULL                        | NULL   | NULL       | NULL      |
| 22 | 8         | 9 Grayhawk Trail    | New York City | NY    | NULL        | NULL       | NULL       | NULL                        | NULL   | NULL       | NULL      |
| 23 | 9         | 16862 Westend Hill  | Knoxville     | TN    | NULL        | NULL       | NULL       | NULL                        | NULL   | NULL       | NULL      |
| 24 | 10        | 4 Bluestem Parkway  | Savannah      | GA    | 37270       | Yovonnnda  | Magrann... | Executive Secretary         | 63996  | NULL       | 10        |

---

## Niveau 10 (Sous-requêtes)

### Question 38

Afficher les employés gagnant plus que le salaire moyen.

```
1 SELECT * from employees where salary > (select AVG(salary) from employees)
```

119 % Aucun problème détecté

Résultats Messages

|    | employee_id | first_name | last_name | job_title                   | salary | reports_to | office_id |
|----|-------------|------------|-----------|-----------------------------|--------|------------|-----------|
| 1  | 37851       | Sayer      | Matterson | Statistician III            | 98926  | 37270      | 1         |
| 2  | 40448       | Mindy      | Crissil   | Staff Scientist             | 94860  | 37270      | 1         |
| 3  | 56274       | Keriann    | Alloisi   | VP Marketing                | 110150 | 37270      | 1         |
| 4  | 67009       | North      | de Clerc  | VP Product Management       | 114257 | 37270      | 2         |
| 5  | 67370       | Elladine   | Rising    | Social Worker               | 96767  | 37270      | 2         |
| 6  | 72540       | Guthrey    | Iacopetti | Office Assistant I          | 117690 | 37270      | 3         |
| 7  | 72913       | Kass       | Hefferan  | Computer Systems Analyst IV | 96401  | 37270      | 3         |
| 8  | 76196       | Mirilla    | Janowski  | Cost Accountant             | 119241 | 37270      | 3         |
| 9  | 84791       | Hazel      | Tarbert   | General Manager             | 93760  | 37270      | 4         |
| 10 | 95213       | Cole       | Kesterton | Pharmacist                  | 86119  | 37270      | 4         |
| 11 | 115357      | Ivy        | Fearey    | Structural Engineer         | 92710  | 37270      | 5         |

---

### Question 39

Afficher le produit le plus cher.

```
1 select * from products where unit_price = (select unit_price from products
2 order by unit_price desc OFFSET 0 ROWS FETCH NEXT 1 ROWS ONLY)
```

119 % Aucun problème détecté

Résultats Messages

|   | product_id | name                      | quantity_in_stock | unit_price |
|---|------------|---------------------------|-------------------|------------|
| 1 | 2          | Pork - Bacon,back Peameal | 49                | 4.65       |

---

### Question 40

Afficher les clients ayant dépensé plus que la moyenne.

```

1  SELECT c.customer_id, c.first_name, SUM(oi.quantity * oi.unit_price) AS total_depense
2  FROM customers c
3  INNER JOIN orders o ON c.customer_id = o.customer_id
4  INNER JOIN order_items oi ON o.order_id = oi.order_id
5  GROUP BY c.customer_id, c.first_name
6  HAVING SUM(oi.quantity * oi.unit_price) > (
7      SELECT AVG(total_par_client)
8      FROM (
9          SELECT SUM(oi2.quantity * oi2.unit_price) AS total_par_client
10         FROM orders o2
11         INNER JOIN order_items oi2 ON o2.order_id = oi2.order_id
12         GROUP BY o2.customer_id
13     ) AS moyennes
14 );

```

119 % ✔ Aucun problème détecté

Résultats Messages

|   | customer_id | first_name | total_depense |
|---|-------------|------------|---------------|
| 1 | 2           | Ines       | 157.92        |
| 2 | 6           | Elka       | 113.58        |
| 3 | 10          | Levy       | 117.41        |

## Question 41

Afficher les employés travaillant dans le même bureau que l'employé **Yovonnda Magrannell**.

```

1  select * from employees where office_id = 10
2

```

119 % ✔ Aucun problème détecté

Résultats Messages

|   | employee_id | first_name | last_name  | job_title           | salary | reports_to | office_id |
|---|-------------|------------|------------|---------------------|--------|------------|-----------|
| 1 | 37270       | Yovonnda   | Magrannell | Executive Secretary | 63996  | NULL       | 10        |

## Niveau 11 (INSERT)

### Question 42

Ajouter un nouveau produit.

```
1 INSERT INTO products VALUES
2 ('Fourchette en plastique', 120, 0.21)|
```

119 %  Aucun problème détecté

Messages

(1 ligne affectée)

### Question 43

Ajouter un nouveau client.

```
1 INSERT INTO customers VALUES
2 ('Ousmane', 'Sow', '1997-03-28', '77-590-7133', 'Avenue Aynina Fall', 'Thiès', 'TH', 2273)|
```

119 %  Aucun problème détecté

Messages

(1 ligne affectée)

### Question 44

Ajouter un nouvel employé.

```
1 INSERT INTO employees (employee_id, first_name, last_name, job_title, salary, reports_to, office_id) VALUES
2 (37272, 'Ousmane', 'Sow', 'Data scientist', 63996, 37270, 10)|
```

119 %  Aucun problème détecté

Messages

(1 ligne affectée)

Lig.: 2, Car.: 56 TABS CRI

## Niveau 12 (UPDATE)

### Question 45

Augmenter de 10 % le salaire des employés du bureau 2.

```
1 UPDATE employees
2 SET salary = salary * 1.10
3 WHERE office_id = 2;
```

119 %

✓ Aucun problème détecté

Messages

(4 lignes affectées)

### Question 46

Ajouter 500 points aux clients de Californie.

```
1 UPDATE customers
2 SET points = points + 500
3 WHERE state = 'CA';
```

119 %

✓ Aucun problème détecté

Messages

(1 ligne affectée)

### Question 47

Modifier le prix d'un produit.

```
1 UPDATE products
2 SET unit_price = unit_price * 1.10
3 WHERE product_id = 3;
```

119 %

✓ Aucun problème détecté

Messages

(1 ligne affectée)

## Niveau 13 (DELETE)

### Question 48

Supprimer un produit.

```
1 DELETE FROM order_items WHERE product_id = 4;
2 DELETE FROM products
3 WHERE product_id = 4;
```

119 % Aucun problème détecté

Messages

(2 lignes affectées)

(1 ligne affectée)

### Question 49

Supprimer un client sans facture.

```
1 DELETE FROM customers
2 WHERE customer_id NOT IN (
3     SELECT DISTINCT customer_id
4     FROM orders
5 );
```

119 % Aucun problème détecté

Messages

(5 lignes affectées)

### Question 50

Supprimer un employé qui n'est le manager de personne.

```
1 DELETE FROM employees
2 WHERE employee_id = 56274 AND employee_id NOT IN (
3     SELECT DISTINCT reports_to
4     FROM employees
5     WHERE reports_to IS NOT NULL
6 );
```

119 % Aucun problème détecté

Messages

(1 ligne affectée)

## Partie III : Défis (Bonus)

### Défi 1

Calculer le chiffre d'affaires total réalisé par chaque client.

```
SQLQuery4.sq...R\Dell (91))* SQLQuery3.s...Dell (86))* create-db-sto...GR\Dell (87)) SQLQuer
```

```
1 SELECT
2     c.customer_id,
3     c.first_name,
4     c.last_name,
5     SUM(oi.quantity * oi.unit_price) AS chiffre_affaires_total
6 FROM customers c
7 INNER JOIN orders o ON c.customer_id = o.customer_id
8 INNER JOIN order_items oi ON o.order_id = oi.order_id
9 GROUP BY c.customer_id, c.first_name, c.last_name
10 ORDER BY chiffre_affaires_total DESC;
```

119 % Aucun problème détecté

Résultats Messages

|   | customer_id | first_name | last_name  | chiffre_affaires_total |
|---|-------------|------------|------------|------------------------|
| 1 | 2           | Ines       | Brushfield | 157.92                 |
| 2 | 10          | Levy       | Mynett     | 117.41                 |
| 3 | 6           | Elka       | Twiddell   | 98.62                  |
| 4 | 8           | Thacher    | Naseby     | 91.20                  |
| 5 | 5           | Clemmie    | Betchley   | 60.73                  |
| 6 | 7           | Ilene      | Dowson     | 24.08                  |

### Défi 2

Trouver le client ayant effectué le plus de commandes.

```
1  SELECT TOP 1
2      c.customer_id,
3      c.first_name,
4      c.last_name,
5      COUNT(o.order_id) AS nb_commandes
6  FROM customers c
7  INNER JOIN orders o ON c.customer_id = o.customer_id
8  GROUP BY c.customer_id, c.first_name, c.last_name
9  ORDER BY nb_commandes DESC;
```

119 %    ✓ Aucun problème détecté

Résultats    Messages

|   | customer_id | first_name | last_name  | nb_commandes |
|---|-------------|------------|------------|--------------|
| 1 | 2           | Ines       | Brushfield | 2            |

### Défi 3

Trouver les 5 produits les plus vendus.



```

1  SELECT TOP 5
2      p.product_id,
3      p.name,
4      SUM(oi.quantity) AS quantite_totale_vendue
5  FROM products p
6  INNER JOIN order_items oi ON p.product_id = oi.product_id
7  GROUP BY p.product_id, p.name
8  ORDER BY quantite_totale_vendue DESC;

```

119 %

✓ Aucun problème détecté

Résultats

Messages

|   | product_id | name                      | quantite_totale_vendue |
|---|------------|---------------------------|------------------------|
| 1 | 3          | Lettuce - Romaine, Heart  | 28                     |
| 2 | 1          | Foam Dinner Plate         | 16                     |
| 3 | 9          | Longan                    | 9                      |
| 4 | 2          | Pork - Bacon,back Peameal | 7                      |
| 5 | 6          | Petit Baguette            | 7                      |

## Défi 4

Trouver les employés ayant un salaire supérieur à celui de leur manager.

```

1  SELECT
2      e.employee_id,
3      e.first_name,
4      e.last_name,
5      e.job_title,
6      e.salary AS salaire_employe,
7      m.employee_id AS manager_id,
8      m.first_name AS manager_first_name,
9      m.last_name AS manager_last_name,
10     m.salary AS salaire_manager
11 FROM employees e
12 INNER JOIN employees m ON e.reports_to = m.employee_id
13 WHERE e.salary > m.salary;

```

119 %    Aucun problème détecté

|    | employee_id | first_name | last_name | job_title                   | salaire_employe | manager_id | manager_first_name | manager_last_name | salaire_manager |
|----|-------------|------------|-----------|-----------------------------|-----------------|------------|--------------------|-------------------|-----------------|
| 1  | 37851       | Sayer      | Matterson | Statistician III            | 98926           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 2  | 40448       | Mindy      | Crissil   | Staff Scientist             | 94860           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 3  | 67009       | North      | de Clerc  | VP Product Management       | 125682          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 4  | 67370       | Elladine   | Rising    | Social Worker               | 106443          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 5  | 72540       | Guthrey    | Iacopetti | Office Assistant I          | 117690          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 6  | 72913       | Kass       | Hefferan  | Computer Systems Analyst IV | 96401           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 7  | 76196       | Mirilla    | Janowski  | Cost Accountant             | 119241          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 8  | 80529       | Lynde      | Aronson   | Junior Executive            | 77182           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 9  | 80679       | Mildrid    | Sokale    | Geologist II                | 67987           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 10 | 84791       | Hazel      | Tarbert   | General Manager             | 93760           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 11 | 95213       | Cole       | Kesterton | Pharmacist                  | 86119           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 12 | 98374       | Estrellita | Daleman   | Staff Accountant IV         | 70187           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 13 | 115357      | Ivy        | Fearey    | Structural Engineer         | 92710           | 37270      | Yovonnda           | Magrannell        | 63996           |

## Défi 5

Afficher les commandes dont la valeur totale dépasse 100 \$.

```

1  SELECT
2      o.order_id,
3      o.customer_id,
4      o.order_date,
5      SUM(oi.quantity * oi.unit_price) AS valeur_totale
6 FROM orders o
7 INNER JOIN order_items oi ON o.order_id = oi.order_id
8 GROUP BY o.order_id, o.customer_id, o.order_date
9 HAVING SUM(oi.quantity * oi.unit_price) > 100
10 ORDER BY valeur_totale DESC;

```

119 %    Aucun problème détecté

| order_id | customer_id | order_date | valeur_totale |
|----------|-------------|------------|---------------|
|----------|-------------|------------|---------------|

## Défi 6

Afficher les bureaux classés selon leur masse salariale totale.

```
1  SELECT
2      e.employee_id,
3      e.first_name,
4      e.last_name,
5      e.job_title,
6      e.salary AS salaire_employe,
7      m.employee_id AS manager_id,
8      m.first_name AS manager_first_name,
9      m.last_name AS manager_last_name,
10     m.salary AS salaire_manager
11 FROM employees e
12 INNER JOIN employees m ON e.reports_to = m.employee_id
13 WHERE e.salary > m.salary;
```

119 % Aucun problème détecté

Résultats Messages

|    | employee_id | first_name | last_name | job_title                   | salaire_employe | manager_id | manager_first_name | manager_last_name | salaire_manager |
|----|-------------|------------|-----------|-----------------------------|-----------------|------------|--------------------|-------------------|-----------------|
| 1  | 37851       | Sayer      | Matterson | Statistician III            | 98926           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 2  | 40448       | Mindy      | Crissil   | Staff Scientist             | 94860           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 3  | 67009       | North      | de Clerc  | VP Product Management       | 125682          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 4  | 67370       | Elladine   | Rising    | Social Worker               | 106443          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 5  | 72540       | Guthrey    | Iacopetti | Office Assistant I          | 117690          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 6  | 72913       | Kass       | Hefferan  | Computer Systems Analyst IV | 96401           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 7  | 76196       | Mirilla    | Janowski  | Cost Accountant             | 119241          | 37270      | Yovonnda           | Magrannell        | 63996           |
| 8  | 80529       | Lynde      | Aronson   | Junior Executive            | 77182           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 9  | 80679       | Mildrid    | Sokale    | Geologist II                | 67987           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 10 | 84791       | Hazel      | Tarbert   | General Manager             | 93760           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 11 | 95213       | Cole       | Kesterton | Pharmacist                  | 86119           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 12 | 98374       | Estrellita | Daleman   | Staff Accountant IV         | 70187           | 37270      | Yovonnda           | Magrannell        | 63996           |
| 13 | 115357      | Ivy        | Fearey    | Structural Engineer         | 92710           | 37270      | Yovonnda           | Magrannell        | 63996           |

## Défi 7

Calculer le panier moyen des commandes.

```
1 SELECT
2     AVG(valeur_totale) AS panier_moyen
3 FROM (
4     SELECT
5         o.order_id,
6         SUM(oi.quantity * oi.unit_price) AS valeur_totale
7     FROM orders o
8     INNER JOIN order_items oi ON o.order_id = oi.order_id
9     GROUP BY o.order_id
10 ) AS total_par_commande;
```

119 % Aucun problème détecté

Résultats Messages

|   | panier_moyen |
|---|--------------|
| 1 | 61.106666    |

## Défi 8

Trouver les clients qui n'ont jamais payé une facture.

```
1 SELECT
2     c.client_id,
3     c.name,
4     c.phone
5 FROM clients c
6 LEFT JOIN invoices i ON c.client_id = i.client_id
7 WHERE i.invoice_id IS NULL;
```

119 % Aucun problème détecté

Résultats Messages

|   | client_id | name   | phone        |
|---|-----------|--------|--------------|
| 1 | 4         | Kwideo | 254-750-0784 |

## Défi 9

Afficher le montant restant à payer pour chaque facture (`invoice_total` - `payment_total`).

```

1  SELECT
2      invoice_id,
3      client_id,
4      invoice_date,
5      invoice_total,
6      payment_total,
7      (invoice_total - payment_total) AS montant_restant
8  FROM invoices
9  ORDER BY montant_restant DESC;

```

119 % Aucun problème détecté

Résultats Messages

|    | invoice_id | client_id | invoice_date | invoice_total | payment_total | montant_restant |
|----|------------|-----------|--------------|---------------|---------------|-----------------|
| 1  | 8          | 1         | 2019-05-20   | 189.12        | 0.00          | 189.12          |
| 2  | 9          | 5         | 2019-07-09   | 172.17        | 0.00          | 172.17          |
| 3  | 5          | 5         | 2019-07-18   | 169.36        | 0.00          | 169.36          |
| 4  | 2          | 5         | 2019-06-11   | 175.32        | 8.18          | 167.14          |
| 5  | 16         | 1         | 2019-03-30   | 162.02        | 0.00          | 162.02          |
| 6  | 10         | 1         | 2019-06-30   | 159.50        | 0.00          | 159.50          |
| 7  | 4          | 3         | 2019-03-08   | 152.21        | 0.00          | 152.21          |
| 8  | 3          | 5         | 2019-07-31   | 147.99        | 0.00          | 147.99          |
| 9  | 18         | 5         | 2019-05-23   | 180.17        | 42.77         | 137.40          |
| 10 | 19         | 1         | 2019-11-23   | 134.47        | 0.00          | 134.47          |
| 11 | 7          | 3         | 2019-09-04   | 133.87        | 0.00          | 133.87          |
| 12 | 11         | 3         | 2019-01-07   | 126.15        | 0.03          | 126.12          |
| 13 | 1          | 2         | 2019-03-09   | 101.79        | 0.00          | 101.79          |
| 14 | 15         | 3         | 2019-11-25   | 167.29        | 80.31         | 86.98           |
| 15 | 6          | 1         | 2019-01-29   | 157.78        | 74.55         | 83.23           |

## Défi 10

Créer une vue SQL nommée `top_customers` qui affiche les clients dont les achats dépassent 500 \$.

```

1  CREATE VIEW top_customers AS
2  SELECT
3      c.customer_id,
4      c.first_name,
5      c.last_name,
6      SUM(oi.quantity * oi.unit_price) AS total_achats
7  FROM customers c
8  INNER JOIN orders o ON c.customer_id = o.customer_id
9  INNER JOIN order_items oi ON o.order_id = oi.order_id
10 GROUP BY c.customer_id, c.first_name, c.last_name
11 HAVING SUM(oi.quantity * oi.unit_price) > 500;

```

119 % Aucun problème détecté

Messages

Commandes réussies.

## Objectifs pédagogiques

À la fin de ce devoir, l'étudiant devra être capable de :

- Concevoir et comprendre un schéma relationnel.
- Manipuler les données avec **SELECT**, **INSERT**, **UPDATE** et **DELETE**.
- Utiliser les clauses **WHERE**, **ORDER BY**, **GROUP BY**, **HAVING** et **LIMIT**.
- Réaliser des jointures (**INNER JOIN**, **LEFT JOIN**) et des auto-jointures.
- Écrire des sous-requêtes simples et avancées.
- Exploiter les fonctions d'agrégation (**COUNT**, **SUM**, **AVG**, **MIN**, **MAX**).
- Interpréter les relations entre les tables et produire des requêtes répondant à des besoins métier.

Ce devoir est progressif, allant des requêtes de base jusqu'à des exercices avancés proches de situations rencontrées en entreprise.